An Adjoint-Strictly-Singular Subcase of the Dual Problem

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Status

This packet records a `partial_result_likely_valid` for Problem 2(iii) in Argyros–Beanland–Motakis [1]. The result is not a full solution of the dual problem; it handles the subcase in which the dual operators have strictly singular adjoints on the reflexive predual.

Source Problem

The source paper constructs, for each $n \in \mathbb{N}$, a reflexive Tsirelson-like space $\mathfrak{X}_{0,1}^n$ such that for every infinite-dimensional subspace Y of $\mathfrak{X}_{0,1}^n$, every product of $n+1$ strictly singular operators on Y is compact. In its final "Problems and Questions" section, the paper asks:

> Does $\mathfrak{X}_{0,1}^{n*}$ satisfy that whenever $S_1, \ldots, S_{n+1} : \mathfrak{X}_{0,1}^{n*} \to \mathfrak{X}_{0,1}^{n*}$ are strictly singular, then the composition $S_1 \cdots S_{n+1}$ is compact, as in $\mathfrak{X}_{0,1}^n$?

The question appears on PDF page 43 of [1]; the local source lines are approximately `source.tex:3867--3874`.

(i) If $\{x_k\}_{k\in\mathbb{N}}$ generates a spreading model equivalent to c_0, $F_k \in \mathcal{S}_1$ for $k \in \mathbb{N}$ and $y_k = \sum_{i\in F_k} x_i$, then a subsequence of $\{y_k\}_{k\in\mathbb{N}}$ generates an ℓ_1^ω spreading model.

(ii) Suppose $\{x_k\}_{k\in\mathbb{N}}$ generates an ℓ_1^ω spreading model, $F_k \in \mathcal{S}_\omega$ and F_k is maximal $\mathcal{S}_\omega$ for each $k \in \mathbb{N}$ (i.e. maximal in $\mathcal{S}_{\min F_k}$). Let $w_k = \sum_{j\in F_k} c_j x_i$ where w_k is $(\min F_k, 3/\min F_k)$ s.c.c. Then a subsequence of $\{w_k\}_{k\in\mathbb{N}}$ generates a c_0 spreading model.

Proof. The proof of (i) is identical to that of Proposition 3.14.

To prove (ii) it suffices to show $\alpha_{<\omega}(\{w_k\}) = 0$. We use Proposition 5.18. Let $\varepsilon > 0$. Find $j_0 > 2/\varepsilon$. Let $j \geq j_0$ and let $k_j \in \mathbb{N}$ such that $36/\min F_{k_j} < \varepsilon$. Let $k \geq k_j$, $\{\alpha_q\}_{q=1}^d$ be $\mathcal{S}_j$-admissible and very fast growing sequence of α-averages such that $s(\alpha_q) > j_0$ for $q = 1, \dots, d$. Clearly, $j < F_{k_j}$. Using Lemma 3.4

$$\sum_{q=1}^{d} |\alpha_q(\sum_{j\in F_k} c_j^{F_k} x_j)| < \frac{1}{s(\alpha_1)} + 6\frac{3}{\min F_k} < \varepsilon.$$

□

Problems and Questions

There are some questions and problems concerning the structure of $\mathfrak{X}_{0,1}^n$ and its dual which are open for us.

Problem 1: (i) Is $\mathfrak{X}_{0,1}^n$ minimal?

(ii) Does any sequence generating a c_0 spreading model have a subsequence equivalent to some subsequence of the basis?

If this is true, then Proposition 3.18 yields that $\mathfrak{X}_{0,1}^n$ is sequentially minimal.

In particular, it is open to us whether two subsequences $\{e_{i_m}\}_{m\in\mathbb{N}}, \{e_{j_m}\}_{m\in\mathbb{N}}$ of the basis, such that $i_m < j_{m+1}$ and $j_m < i_{m+1}$ for all $m \in \mathbb{N}$, are equivalent.

Moreover, we do not know which class of Banach spaces in the classification appearing in [14] the subspaces of $\mathfrak{X}_{0,1}^n$ belong to.

The next problem concerns the structure of $\mathfrak{X}_{0,1}^{n*}$ and its strictly singular operators.

Problem 2: (i) Does any block sequence in $\mathfrak{X}_{0,1}^{n*}$ contain a subsequence generating a c_0^k, $k = 1, \dots, n$ or ℓ_1 spreading model?

(ii) Does any subspace of $\mathfrak{X}_{0,1}^{n*}$ admit $c_0^k, k = 1, \dots, n$ and ℓ_1 spreading models?

The latter is equivalent to the corresponding problem for quotients of $\mathfrak{X}_{0,1}^n$, namely if every quotient of $\mathfrak{X}_{0,1}^n$ admits c_0 and ℓ_1^k, $k = 1, \dots, n$ spreading models. Note that Cor. 3.20 yields that the same question for quotients of $\mathfrak{X}_{0,1}^{n*}$ has an affirmative answer.

Proof Intuition

The source theorem already proves the desired product-compactness property on the predual $\mathfrak{X}^n_{0,1}$. Since $\mathfrak{X}^n_{0,1}$ is reflexive, every operator on ${\mathfrak{X}^n_{0,1}}^*$ has an adjoint that can be identified with an operator on $\mathfrak{X}^n_{0,1}$. Therefore the dual problem would follow immediately for any family whose adjoints are strictly singular on $\mathfrak{X}^n_{0,1}$: apply the source theorem to the adjoints, then use Schauder's theorem to pass compactness back to the original product.

This isolates the missing point in the full problem. Strict singularity is not self-dual for arbitrary Banach spaces, so one cannot simply assume that a strictly singular operator on ${\mathfrak{X}^n_{0,1}}^*$ has strictly singular adjoint on $\mathfrak{X}^n_{0,1}$.

The Partial Result

Theorem 1. *Let $n \in \mathbb{N}$ and let $\mathfrak{X}^n_{0,1} = \mathfrak{X}^n_{0,1}$ be the reflexive space constructed in [1]. Let $S_1, \ldots, S_{n+1} \in \mathcal{L}({\mathfrak{X}^n_{0,1}}^*)$. Assume that, after identifying ${\mathfrak{X}^n_{0,1}}^{**}$ with $\mathfrak{X}^n_{0,1}$, each adjoint*

$$S_i^* : \mathfrak{X}^n_{0,1} \longrightarrow \mathfrak{X}^n_{0,1}$$

is strictly singular. Then $S_1 S_2 \cdots S_{n+1}$ is compact on ${\mathfrak{X}^n_{0,1}}^*$.

Proof. By the main theorem of [1], applied to the subspace $Y = \mathfrak{X}^n_{0,1}$, the composition of any $n+1$ strictly singular operators on $\mathfrak{X}^n_{0,1}$ is compact. Since each S_i^* is strictly singular by hypothesis, the operator

$$S_{n+1}^* S_n^* \cdots S_1^* : \mathfrak{X}^n_{0,1} \longrightarrow \mathfrak{X}^n_{0,1}$$

is compact.

The adjoint of the original product is

$$(S_1 S_2 \cdots S_{n+1})^* = S_{n+1}^* S_n^* \cdots S_1^*,$$

and hence is compact. By Schauder's theorem, a bounded operator is compact if and only if its adjoint is compact. Therefore $S_1 S_2 \cdots S_{n+1}$ is compact on ${\mathfrak{X}^n_{0,1}}^*$. □

Equivalent Reduction

Because $\mathfrak{X}^n_{0,1}$ is reflexive, every bounded operator on ${\mathfrak{X}^n_{0,1}}^*$ has a preadjoint on $\mathfrak{X}^n_{0,1}$: namely, for $S \in \mathcal{L}({\mathfrak{X}^n_{0,1}}^*)$, identify S^* as an operator from ${\mathfrak{X}^n_{0,1}}^{**}$ to ${\mathfrak{X}^n_{0,1}}^{**}$, hence from $\mathfrak{X}^n_{0,1}$ to $\mathfrak{X}^n_{0,1}$. Thus Problem 2(iii) would be solved if one could prove the implication

$$S \in \mathcal{S}({\mathfrak{X}^n_{0,1}}^*) \quad \Longrightarrow \quad S^* \in \mathcal{S}(\mathfrak{X}^n_{0,1}).$$

The theorem above proves the dual product conclusion exactly under this additional adjoint-strict-singularity hypothesis.

Limitations and Novelty Check

This packet does not prove the implication $S \in \mathcal{S}({\mathfrak{X}^n_{0,1}}^*) \Rightarrow S^* \in \mathcal{S}(\mathfrak{X}^n_{0,1})$. That implication is false for general Banach spaces, so it would require a special property of $\mathfrak{X}^n_{0,1}$ or its dual.

Local run indexes were searched for arXiv:1309.4516, the paper title, "dual strictly singular", and adjoint/strictly singular keywords. No duplicate packet was found. Web searches on June 15, 2026 found the source paper, the related higher-order distortion paper arXiv:1408.5065, and the related asymptotic-symmetry paper arXiv:1902.10098, but no later full answer to Problem 2(iii).

Human Review Recommendation

Review as a small partial subcase/reduction. The proof itself is short; the main point to verify is that the statement is not overread as a full solution. The remaining full problem is whether strict singularity on ${\mathfrak{X}^n_{0,1}}^*$ forces strict singularity of the adjoint on $\mathfrak{X}^n_{0,1}$ for this particular reflexive Tsirelson-like space.

References

[1] S. A. Argyros, K. Beanland, and P. Motakis, *Strictly singular operators in Tsirelson like spaces*, arXiv:1309.4516.

[2] J. Schauder's compactness theorem: a bounded operator between Banach spaces is compact if and only if its adjoint is compact.